The a_0 Law with Cold Dark Energy Counted Once: the Form the Record Closes On, Its Gates, and What Stays Open

Working note (PAPER72), version 1.0 draft, 2026-11-09, not deposited. AI-assisted research programme; not peer reviewed; not reproduced by any outside group. Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (v2.1, DOI 10.5281/zenodo.24092073, field equations and the record since) and PAPER71 (v1.1, DOI 10.5281/zenodo.24095262, the switch class). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

This note states the form on which the record closes and grades it. It does not complete the theory. The a_0 law alone fails the CMB, the growth of structure, clusters and the Bullet, each for want of the same cold, clustering component of about 5.4 times the baryons. Every way of keeping that component out of galaxies has a failing verdict; the last one is recorded here. The form chosen by the owner on 2026-11-09 counts the component once: it is in galaxies too, and around condensed baryons the law describes how it sits. It is real mass, not a second force. In the adopted action this is the static state's own reading: the law's halo is the khronon sector's energy density, carried by the one conserved charge that is dust in the expanding background. We regrade the form with criteria frozen before any number. On groups and clusters it is not falsified, by a gate that can fail from one side only, and the richest clusters sit at its ceiling; the Bullet passes only if the subcluster's stripped gas is counted. On hosted dwarfs it fails: it removes most of the law's shortfall in every class, but two classes stay high and four low rows cannot be reached. On KiDS lensing by galaxy type the test returns no verdict; its readings put early types between the law and the form's truncated halo and cannot place discs; on host-free lenses the form passes a gate that does not separate it from the law. The form passes rotation curves. Its CMB, lensing and growth rows hold when computed, with the amplitude of fluctuations unsolved. No solution has been computed that takes infalling dust to the static state; that state is unstable in the one limit analysed; no single dust term serves the CMB and galaxies; and no cluster state has been solved. We list what is open, the measurement that would decide each item, and our own wrong estimates.

1 Two forms

Cold dark energy is this record's name for a gravitating field component with $w \approx 0$ at the cosmic amount; it is not a particle. Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are carried throughout and never pooled; pairs of numbers are canonical/alt unless a row says otherwise.

Form A, the law alone, is what PAPER70 states. It passes rotation curves and the Solar System. It fails the CMB (which needs a cold component of 5.41 times the baryons), growth, X-COP by $+0.4675/+0.4345$ dex, the Bullet ($Z = -4.55$, the mass sits on the gas) and Coma's ultra-diffuse galaxies ($+1.12$ dex with a static external field, $+1.41$ with a moving host).

Form B, the law with cold dark energy counted once, is the single-counting reading that lanes PD1 and PD2 graded with one fixed share per halo. As graded there it is FAIL/FAIL: it passes its population budget, rotation curves (rms 0.0563/0.0558 against a line of 0.0660) and the carried cosmology, and fails KiDS's pooled outer points (worst -0.635 dex), the groups and the Bullet's amount; X-COP passes only in a reported bracket ($+0.019/+0.010$).

On 2026-11-09 the owner chose B as the form the record closes on, with both forms scored side by side from the record. The choice is of a form, not of a pass: no label was changed by it. It reopens the cold sector retired on 2026-11-01 for this one reading. Section 3 is the regrade that followed.

2 Form B in the adopted action

The action of PAPER70 has certified field equations (lane QX17). A note written after the decision, and checked independently for this paper, reads form B out of them for a static clock.

- **The law's halo is energy density.** The khronon sector's energy density contains $\Pi = D\cdot P$, and the static weak-field 00 equation is Poisson's with source $\rho_b + D\cdot P$. Light and matter both see it.
- **One conserved charge.** The khronon equation is a conserved current whose charge is, to leading order, $D\cdot P + \mu_K^2(Q-1)/K$. With $P = 0$ it is cold dust in the expanding background; around a static source it is the law's halo plus the static dust term $-\mu_K^2\Phi/K$, which is zero at $\mu_K = 0$. Nothing is counted twice.

- **The static state is in force balance.** With $J(g) = (a_0^2/K)[u - \ln(1+u)]$, $u = g/a_0$, the law sector's stress satisfies $\partial_j \Pi_{ij} = (D \cdot P)\, g_i$. In the deep field $J = g^2/2K$, which is the pressure of an isothermal halo with $\sigma^2 = \frac{1}{2}\sqrt{G M_b a_0}$: the relation PD1 had to assume. The identity holds for any $J(g)$ with a curl-free field, so it shows that the action's static state balances itself; it is not evidence for this J.
- **Force balance is not stability.** With the metric frozen (the decoupling limit), lane TPA1 finds the action's law-on phase gradient-unstable, with no ghost, wherever the clock's own dust density is below a critical value. For a point mass that value is 0.08 to 0.9 of the law's phantom density, and the e-folding times on its grid are 5.0 Myr to 0.62 Gyr. TPA1 states this for its law-on phase. The note on form B reads the form's static state, which has almost no dust of its own, as lying in that range. The analysis coupled to gravity has not been done.

What the action does not give is stated in Section 5, items 1 to 4.

3 The regrade

Each lane froze its criteria in a commit before any graded number was computed (KO2 ran a model-only design toy before its freeze and says so), and has known answers that must pass and mutation twins that must fail. Each was re-run by a session other than the one that wrote it, and its labels reproduced; the re-runs are listed in `closure_2026_11_09/RERUNS_2026-11-09.md` and `fable52_push/README.md`. The lanes relabel nothing.

3.1 Groups, clusters and the Bullet (RB1): NOT FALSIFIED (ONE-SIDED) on both footings

In form B a hot system holds $R = 5.41$ times the baryons that fell into it, and the component stays when gas is expelled or stripped. The retained fraction $f_{\rm ret} = R\, M_{b,\rm obs}/(M_{\rm dyn} - M_{b,\rm obs})$ can therefore not exceed 1 at a system's edge. This gate can fail from above only; the two-sided amount gate stays FAIL as PD1 graded it.

- **Groups.** The 20 hydrostatic groups read $f_{\rm ret} = 0.597$ [0.556, 0.637] at R_{500}, none above 1. Form B reads the missing 40 % as gas outside R_{500}. Nothing on disk measures it.
- **Clusters sit at the ceiling.** X-COP at its last radius reads 1.095 [0.885, 1.259], with 4 of 7 clusters above 1. At R_{500} on the forward-fitted mass the class reads 1.057 [1.046, 1.072], above the ceiling beyond its errors; this row is a reported bracket, and graded there the gate would fail. On the NFW-fit hydrostatic mass the same class reads 0.959. The verdict for the deepest wells turns on about 10 % in the hydrostatic mass.
- **Trend.** $f_{\rm ret}$ rises with depth: slope +0.160 [+0.080, +0.225] in $\log M_{500}$ from groups to X-COP.
- **Radial shape.** One collisionless component with the record's own profile, anchored at each cluster's last radius, gives missing-mass fractions 0.72 / 0.63 / 0.55 / 0.47 / 0.35 per r/R_{500} bin against the measured 0.71 / 0.66 / 0.58 / 0.46 / 0.32 (canonical); bin residuals −0.007 / +0.033 / +0.006 / −0.030 / −0.041 dex against a line of 0.10. This label rests on a post hoc amendment; under the frozen known answer the gate is NO VERDICT.
- **The Bullet.** PD1's 2.6σ deficit in the subcluster is the bookkeeping of the stripped gas. With the gas shared in proportion to the stars the amounts read −0.45 / +0.37 (main / sub) and pass; with nothing reassigned they read +0.49 / −2.58 and fail. No measurement on disk fixes the subcluster's gas before the merger.

3.2 Hosted dwarfs (RB2): B FAILS ON THE HOSTED ROWS on both footings

In form B a dwarf keeps the phantom of its own baryons inside a host, cut only by tides. There is no external field effect.

- **What it removes.** Coma's ultra-diffuse galaxies move from +1.120 to +0.326 dex (alt +0.283) and read B CONSISTENT; the interval contains zero only with the systematics budget, and on statistics alone it is [+0.105, +0.547]. The Milky Way's classical dwarfs read B CONSISTENT as a class (+0.281/+0.241), with Draco, Ursa Minor and Sextans high row by row.
- **Two classes fail high.** The Milky Way's ultra-faints sit at +0.634 [+0.433, +0.865] (alt +0.592) and M31's bright dwarfs at +0.273 [+0.085, +0.473] (alt +0.231). The statistic carries measurement scatter only; a full systematics budget was not computed, since that would be a change after the freeze.
- **Rows that cut the other way.** NGC 1052-DF2 and DF4 sit 0.74 and 0.94 dex below B and at Newton on their baryons. Tides reach them only if NGC 1052 holds 17 and about 100 times the mass B gives it inside

their projected distances. NGC 147 and IC 1613 sit 0.56 and 0.54 dex below B; the law with its external field leaves them as low. Crater II is the one low row tides can reach.

3.3 Lensing by galaxy type (KO2): NO VERDICT for late and early types on both footings

All 181,477 KiDS lenses were fitted with every catalogued neighbour modelled off-centre (888,795 pool objects), in place of a veto. The statistic is the ratio of the amplitude beyond the inner window to the inner amplitude; the law as a galaxy's own field gives 1 and no extended field gives about 0.

- **Why no verdict.** One of 13 known answers failed (a cross-shear pull of 3.11 among 15 amplitudes). Without it the frozen bracket rule gives NOT SEPARATED for both types. One mutation twin did not bite: the ratio measures the shape of the outer profile, not its level.
- **Early types lie between.** They read 0.487 ± 0.071 in the primary design and 0.739 ± 0.115 with lens and neighbour amplitudes untied: below the law's level in every design and above form B's truncated halo (0.038/0.007 in the primary design).
- **Discs cannot be placed.** They read 0.277 ± 0.183 tied and 1.071 ± 0.199 untied. Form B planted with the measured cross-shear as noise reads 1.030 ± 0.282 untied, so the data cannot tell the law from form B for discs.
- **An earlier lead does not generalise.** Lane KO1, and PAPER71 beside its gate, reported that host-free late types show no outer signal (-0.19 ± 0.39 against 1.34 ± 0.18 for early types). Both labelled it a lead and not a result: seen after the freeze and sharing the veto's edge. Plain stacks of all lenses are flat for both types (late outer-to-inner ratio 0.93 ± 0.13, early 1.09 ± 0.08), though they include the neighbours' signal. With neighbours modelled, the deficit appears only inside the host-free subset (-0.03 ± 0.28), which carries the veto's edge.

A look at the host-free lenses, committed before KO2 had a lensing number, compares form B at PD1's share with KO1's shape ratio. It is post hoc and models no neighbours. At the frozen veto radius form B reads 0.510 against a measured 0.607 ± 0.094, with the law as the lens's own field 4.2σ off on the other side. At the widest veto, where the edge effect is weakest, form B reads 0.525 against 0.811 ± 0.134, 2.1σ low, with the law 1.4σ off. The note read from it that one share does not fit late and early types together; ONE1's frozen type-difference line passes at every veto, so the graded statistic does not read that. The same note predicted blind that KO2's late-type ratio would be at most about 0.5; the tied design meets that (0.277 ± 0.183) and the untied one does not (1.071 ± 0.199), so it is not confirmed. Lane ONE1 then graded form B on those lenses with every catalogued neighbour carrying the same share: KiDS host-free PASSES/PASSES by its frozen gate at the widest veto, for PD1's share and for that row only. The shape ratio there is 0.811 against a model of 1.003 (-1.14σ; alt 0.809 against 0.847); late types read 0.084 against 0.792 (-1.46σ) and early types 0.882 against 0.999. At the frozen veto radius, which carries the edge, the model misses (0.607 against 0.924, -2.68σ). The law with no end gives 0.98 at the widest veto, also inside 2σ, so the pass does not separate form B from the law, and the late-type error there (0.42) is too large to test discs. The look's 0.510 to 0.525 was mainly a matter of calibration: with the inner calibration fitted, the model reads 0.86/0.73 at the widest veto without neighbours and 1.00/0.85 with them. [PENDING: ONE1's KiDS script re-run by its lead session]

3.4 The last way of keeping the component out (HE1): FAILS on both footings

HE1 held the galaxy's share outside a wall as an envelope in pressure balance with the law's field. The envelope exists, is stable and fills 0.2 % of space or less. It fails KiDS's late-type stacks: none of its three stated shares, at any outer radius, passes its first four gates together. Post hoc and with low power, at the widest veto the record's share fits those stacks (p 0.73/0.83), as does the law. Its fifth gate fails as frozen on two clusters near R_{500} only, and the label does not rest on it. Re-run by its lead session: both labels and every entry of its result file reproduce; only timing lines differ. Lane TPA1 wrote out and certified the covariant equations of the two-phase action (EQUATIONS COMPLETE; health UNHEALTHY): its switch wall and its dust phase are healthy, and its law-on phase has the gradient instability of Section 2. With PAPER71's closed switch class, no reading that keeps the component out of galaxies passes the record's gates.

3.5 The cosmology rows, computed (CO1): ROWS HOLD on the amended reading, with an inherited halo list

The decision listed form B's cosmology rows as carried from the cold reference. A further lane computed them. Form B's linear content passes the primary CMB on raw Planck PR4 and ACT DR6 spectra ($-65.21/-65.07$ against a line of 9; cold to baryon 5.410/5.412, fitted). That row replays a fit stored by an earlier lane, and its yardstick is the likelihood package's own test point, not a framework quantity. PR4 lensing reads $0.9617 \pm 0.0231/0.9564 \pm 0.0230$ of the predicted amplitude ($-1.66/-1.89\sigma$ against a 2σ line; shape χ^2 8.56 over 9 bandpowers), and DESI DR1 $f\sigma_8$ passes (largest pull 1.22/1.23). A cold component of the fitted amount gives the cold-component spectra, so these three rows holding is not independent evidence for the framework. For the layout frozen in the criteria (galaxies: the Lagrangian share as the law's phantom; groups and clusters: the share inside R_{200c}, part of it kept in the reference profile) the matter power changes by 0.6/1.0 % at $k = 1$ and 5.1/8.8 % at $k = 3\,h\,\mathrm{Mpc}^{-1}$, and raw KiDS-1000 cannot see it on the 104 points the reference run supports.

- **Two readings of the label.** One mutation arm did not bite as frozen; the label rests on an amendment committed before any graded number. Read as frozen it is NO VERDICT.
- **The halo list is inherited:** a mass function, a bias and a concentration relation calibrated on simulations. The redistribution rows are externally calibrated and say so.
- **The scope decides it.** If a cluster held the whole dust of its turnaround sphere, the power at $k = 1$ would be 2.9 times higher and raw KiDS would reject it. The form's cosmology needs the cluster share tied to the baryons present (5.41 times them) and counted inside R_{200c}; RB1 grades the same bookkeeping at R_{500} or the last measured radius.
- **The amplitude is not solved.** The fit's σ_8 is 0.8405/0.8429, with no low-multipole data to hold it. Raw KiDS fits the record's reference run at 0.811 with $p = 0.014$ and form B at its own amplitude with $p = 0.0015/0.0009$ (a bracket with constant-factor scaling, which understates it). The forest row stays carried. [PENDING: CO2: the amplitude refitted with low-multipole polarisation, label and origin commit]

4 Form B after the regrade

Gate	Reading for form B	Lane
Rotation curves	pass, rms 0.0563/0.0558 (line 0.0660)	PD1
Solar System	pass, carried	PD1
CMB peaks, CMB lensing, growth rate	hold when computed; NO VERDICT with one mutation arm as frozen; redistribution rows on an inherited halo list	F1, CO1
Forest	carried from the cold reference	PD1
Amplitude of fluctuations	not solved	CO1
Clusters, amount	not falsified (one-sided); at the ceiling; two-sided gate FAIL	RB1, PD1
Clusters, radial shape	consistent under a post hoc amendment; NO VERDICT as frozen	RB1
Groups	not falsified (one-sided), f_{ret} 0.597; two-sided gate FAIL	RB1, PD1
Bullet	consistent only if the subcluster's stripped gas is counted	RB1
Coma UDGs	consistent on the systematics budget	RB2
Milky Way classical dwarfs	consistent as a class	RB2
Ultra-faints, M31 bright dwarfs	FAIL high	RB2
DF2, DF4, NGC 147, IC 1613	against the form; tides cannot reach them	RB2
KiDS lensing by type	NO VERDICT; early types between the law and the form	KO2
KiDS host-free lenses	passes at the widest veto with neighbours carrying the share; does not separate the form from the law	ONE1
Outer cluster shear	FAIL, about 3σ with systematics	ZN4, FR1
Static state reached by collapse	not computed	QX17
Stability of the static state	unstable in the decoupling limit; coupled analysis not done	TPA1

5 What stays open

1. **The collapse.** For a static clock the action's equations give the law's shape and hold it. No solution takes a free-falling clock, dust falling in with the baryons, to that state. This is the spiral wall of the 2026-11-01 ledger.
2. **Stability of the static state.** The law-on phase is gradient-unstable in the decoupling limit when the clock carries little dust of its own, which is how the note on form B reads the form's galaxies. The coupled analysis has not been done.
3. **One dust term.** With the quadratic $K(Q)$ the background has $w = w_0/(w_0 + a^3)$ with $w_0 = I_0/4\mu_K^2$. Our arithmetic, not a lane, with H_0 70 and Ω_c 0.26: $w_0 = 1.063 \times 10^{-8}/\mu_K^2$ (μ_K per Mpc), so at $\mu_K = 3.71$ per Mpc the component still has $w = \frac{1}{2}$ at $z = 1089$. Dust by recombination needs μ_K well above that; the galaxy-lensing bound is 0.1.
4. **Clusters are not a static-clock state.** Closing X-COP with the static dust term needs μ_K of 1.78/1.74 per Mpc, 17 to 18 times the galaxy bound. No other state has been solved.
5. **Groups** are untested from below: form B needs 40 % of their baryons to lie outside R_{500}.
6. **The richest clusters** sit at or above the form's ceiling, depending on the hydrostatic mass used.
7. **The Bullet** passes only with gas credited to the subcluster that no measurement fixes.
8. **Ultra-faints and M31's bright dwarfs** sit above the form.
9. **DF2 and DF4** sit below it and need a host 17 to 100 times heavier than the form gives.
10. **KiDS.** On all lenses, early types need more beyond the inner window than PD1's truncated halo and less than the law. On host-free lenses the form passes a gate that the law passes too. Discs are not tested by either.
11. **The outer cluster shear** stands near 3σ.
12. **Cosmology.** The amplitude is not solved, the forest row is carried, and the redistribution rows rest on an inherited halo list and on the cluster share being counted inside R_{200c}.

6 What would decide it

- The spherical collapse of the clock around a baryon mass with the record's own equations. It is a mechanism lane and has not been started.
- The stability of the static state with gravity coupled.
- The baryons between R_{500} and the turnaround radius around the same groups (thermal SZ, X-ray, stars) with a lensing mass to that radius. Form B needs $f_{\rm ret} = 1$ at turnaround; a census that falls short falsifies it.
- A cluster mass beyond R_{500} that does not assume hydrostatic equilibrium.
- A measured mass for NGC 1052 inside 79 and 165 kpc.
- The dwarf classes regraded with a full systematics budget, frozen before the numbers.
- Lensing by type with more area. A spectroscopic veto with the redshifts on disk has no power: the late-type error at the widest veto is 0.68/0.62 of the law in KO1 and is forecast at 0.57/0.52 with such a veto, against about 0.47 needed.
- A CMB fit with low-multipole polarisation, to fix the amplitude that weak lensing then tests.
- Gaia DR4 (2026-12-02), scored as pre-registered.

7 Our own estimates

Recorded before the numbers that tested them. (i) The KO2 brief put the error on the disc ratio near 0.2 and expected discs low and early types at 0.7 to 1.0. A model-only toy at the freeze gave the brief's ratio about twice the error of its replacement (0.37 against 0.19), so the lane replaced the primary statistic before computing; the brief's ratio then measured an error of 0.278 on discs, and discs came out undecided. (ii) The expectation that discs carry no outer field rested on the host-free lead, which does not generalise to the full sample. (iii) The RB1 brief expected X-COP near 0.96; it reads 1.095 on the forward-fitted mass. (iv) RB2 expected M31's bright dwarfs to be borderline; they fail. (v) The Bullet estimate, closer than 2.6σ once the stripped gas is counted, held; so did the group range and the Coma shift.

Method

Lanes KO2, RB1, RB2 and CO1 are under `closure_2026_11_09/`; HE1, ONE1, TPA1 and the notes FB1 and KO3F are under `fable52_push/`. None is registered in the closure record and none relabels an earlier lane. Post hoc items are marked in their lanes: RB1's amendment on the cluster shape gate, KO2's four amendments (the third, form B's prediction, was committed before any fitted amplitude was opened), and KO2's mutation twin that did not bite. RB2 has no amendment. This note claims no Lean theorem of its own; the Lean build that earlier lanes compiled with was removed from the machine on the evening of 2026-11-09 and a build inside the repository has been used since 22:17 that evening, so a Lean file's status is what its lane's compile output says. Distances for the KiDS lenses use the record's fiducial flat background (Ω_m 0.3, H_0 70), as in the staged sums. No halo model, no concentration relation and no simulation fit enters a graded row of KO2, RB1 or RB2; one externally calibrated bracket in RB1 is labelled as such. CO1's redistribution rows inherit a mass function, a bias and a concentration relation calibrated on simulations, and say so. The owner's decision is recorded in `closure_2026_11_09/OWNER_DECISION_2026-11-09_FINAL_FORM_B.md`. The supplement holds every lane file.